Willamette Applicant Trajectory: Predicting Student College Enrollment Destination

DATA 510 M1 Project Proposal

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1. Title & Team Metadata

Project Title: Willamette Applicant Trajectory: Predicting Student College Enrollment Destination

Team Members:

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Owner Product Lead:

Seira Ramchandani

Peer Stakeholder POs:

Mike Kimmell

Dylan Ray

Studio Session: 3

Team Size: 2

2. Problem Framing and Research Questions

Problem Statement:

Willamette University's Admissions Office and enrollment management team invest significant resources in recruiting prospective students, yet a meaningful portion of admitted applicants ultimately choose to enroll elsewhere.

Currently, the Admissions office lacks a data-driven framework for segmenting admitted applicants by their predicted matriculation destination, which means that recruitment and yield strategies must be applied broadly rather than targeted to what differentiates Willamette from the other schools an applicant may be looking at. This limits the effectiveness of outreach efforts at a critical stage in the enrollment funnel. Understanding whether a student is likely to choose a private Northwest institution or a large public university would allow the team to tailor materials, financial aid messaging, and communications to what is most likely to influence a student's decision.

Our project aims to address that gap by building a predictive model trained on the past 3 years of historical applicant data matched to National Student Clearinghouse enrollment data. The model will classify admitted students into destination categories, initially defined as private NW5 institutions and large public universities (e.g., OSU, UO, UC system). NW5 consists of the following 5 schools: Willamette University, Reed College, Lewis & Clark College, Whitman College, and University of Puget Sound. Additional segments, such as community colleges or other private institutions, may be incorporated as informed by exploratory data analysis. Predictions generated by this model will be used by the Admissions Office to segment incoming applicants and enable targeted outreach strategies accordingly.

Research Questions:

1. What applicant characteristics most strongly predict whether an admitted student will enroll at a private NW5 institution or a large public university?
2. Are there additional meaningful enrollment destinations beyond the initial categories that arise from the data, and what applicant profiles define those groups?

Charter Alignment Note:
While the Charter version included appending application types directly to each application in Slate, this integration is pending confirmation of technical feasibility. Currently, we plan for the model output to be delivered in a format that allows the Admissions Office to test and segment applicants by prediction and apply targeted outreach strategies accordingly.

3. Impact and Stakeholders

Who Benefits

Primary Stakeholders
The primary beneficiaries of this project are William Mullen, VP for Enrollment Management, and the broader Admissions team at Willamette University. If the model performs well, the Admissions office gains a practical tool that can assist with a more efficient allocation of recruitment and outreach resources and, ideally, improved yield rates over time.

Indirect Stakeholders
The Willamette Marketing team stands as an indirect beneficiary, as the model insights could inform messaging and campaign strategies. Prospective students could also benefit from a more targeted outreach that contains relevant communication to their needs.

Organizational Impact (plausible, not hype)

A reliable prediction model would give the Willamette Admissions office a replicable, data-informed process that can be refined for each cycle and utilized in yield strategy. Over time, even modest improvements in the enrollment rate can have meaningful financial and institutional implications.

Shared Theme and Peer PO Influence

Our peer POs are matched on *Community institutions and content engagement for outreach*. We expect them to push us on explainability and ease of use of the final deliverable, as well as actionability. Our peer stakeholders' perspectives will shape our priorities in making sure that the insights are realistic and implementable in their existing workflows.

4. Data Sources and Access Plan

Source	Description	Access	Volume (est.)	Refresh	Instructor approval
1. Technolutions Slate CRM	Row-level applicant records	via the Slate Configurable API & Technolutions SFTP server (Note below)	~10,000 records	Historical pull covering a 3-year window. Refresh cadence to be determined based on project needs	**Approved**
2. National Student Clearinghouse (NSC) data	Enrollment destination data for non-enrolled admits	Delivered as CSV files through SFTP server	~10,000 records	Historical pull covering a 3-year window. Refresh cadence to be determined based on project needs	**Approved**

Notes

1. Slate Configurable API documentation: https://knowledge.technolutions.net/docs/data-and-document-management-overview Data access approved by William Mullen. Seira Ramchandani will begin building the data pipeline and API connection with Mike Kimmell early next week. Data exports will be pulled from the /outgoing folder of the SFTP server. Instructor approval has been confirmed in the project's #blockers Discord channel.
2. Data access approved by William Mullen. Instructor approval has been confirmed in the project's #blockers Discord channel.

5. Data Engineering Plan

Repo bootstrapped from the DS3 template:

- data/raw/ : Original files as received from the SFTP server. Gitignored. Never modified.
- data/external/ : Any supplementary reference data not generated by the team (e.g., institution classification lists for defining NW5 or large university categories).

- data/interim/ : Outputs from cleaning, joins, and intermediate transformations.
- data/processed/ : Analysis-ready dataset used for modeling. Committed to the repo if file size permits; otherwise, stored separately and will be documented.
- src/ingest/ : Ingestion scripts for Slate Configurable API
- notebooks/ : Follow the naming convention in template, and reference data from data/interim/ or data/processed.

Pipeline

1. src/ingest/ - Pull raw Slate and NSC data from the SFTP server into data/raw/
2. src/features/ - Clean, join, and build features from data/raw/ into data/processed/
3. src/models/ - Train, evaluate, and run inference on processed data
4. src/viz/ - Generate figures and reporting visualizations for notebooks and deliverables

6. Planned Analysis

Statistical and/or machine learning methods; evaluation strategy (baselines, metrics, validation); what would count as *answering* the question. Each major analytical step should map to a Product Backlog Item with a **Create / Observe / Analyze** triple.

Exploratory Analysis and Category Validation

- Feature engineering
    - Transformation of raw applicant attributes into model-ready features
    - Figuring out which of the features/aspects of a student's profile have the most impact on where they end up attending college
- Chi-Squared test of independence
    - Could help us identify which variables are worth including in the predictive models
- Clustering
    - Can be applied to the dataset to identify natural groupings in enrollment destination patterns.

Predictive Modeling

- Multinomial Logistic Regression (baseline)
    - Well-suited to a multi-class categorical outcome, since we predict there will be a combination of certain features that predict an enrollment outcome
- Support Vector Machines
    - Can be utilized to classify students into the categories of their chosen schools

Success

We will consider this analysis successful if:

- Clustering reveals interpretable groupings that validate or meaningfully refine the existing destination categories

- Feature importance results are interpretable enough for the Admissions team to understand why a student is predicted to go to a certain destination type
- The model performs well enough to be used as a segmentation tool for the team

7. Program Integration (Five Pillars)

Pillar	Integration into Project
Data Engineering	Data Engineering will be integrated into this project through the creation of a data pipeline and an API connection to obtain the data.
Analytics/Machine Learning	Machine Learning and Analytics are a key part of this project. We will use multiple machine learning techniques to evaluate which features of a student profile are significant and could indicate the type of institution that the student will attend. Without a proper understanding of the variables and features used in each algorithm, the end result would be meaningless.
Visualization and Communication	Communicating with non-technical audiences will be essential in this project, as the primary stakeholder is the Vice President of Enrollment Management. We will use clear communication of our methods, including the ethical considerations, to ensure full understanding of the project.
Ethics and Responsible Use	None of the data that will be utilized will contain any Personally Identifiable Information (PII), protecting the identities of the students whose data we will be using. We are also taking into account and will do all we can to ensure the algorithms and analytics that we perform do not rely on stereotypes or reinforce inequities in higher education.
Statistics/Research Design	The structure of this project integrates research design. The project requires a clear layout of what the research questions are, the data that will be utilized, how it will be analyzed, and what it would mean to

	successfully answer the questions. By having a clear roadmap and planning out the actions that will be taken, research design is integrated into this project. Statistics will be integrated into this project through various statistical tests that will be performed on the data to assist in answering the research questions.

8. Ethics and Risk Notes

The main ethical risk is confidentiality and dealing with PII. In order to maintain the privacy of the student data that we will obtain and utilize, no PII will be present in the actual data that we use. If it is used to join on, it will then be deleted before any analysis or training is done. We will also be signing a formal confidentiality agreement, ensuring that students' information will not be exposed to anyone, whether that be other members of the capstone class, stakeholders, or members of the public.

The other ethical consideration is in the use of predictive models. Since the models will utilize data that has been unavoidably influenced by inequities in collegiate recruitment and enrollment, they will be closely evaluated and subsequently altered if necessary to avoid reinforcing any inequities.

9. Timeline and Milestones

Week	Plan
4 (now)	Create backlog items; meet with the primary stakeholder; finish project proposal
5	Work on creating API pull and datasets. There is a potential risk of not being able to access the data or not being able to create the dataset without the PII. **Dependency:** Data access
6	Clean datasets and put them in a format that makes sense for analysis. Perform EDA and obtain all required context to ensure a complete understanding of the data.
7 (M2 Data Summary)	Begin machine learning and statistical testing. Ensure that all planned approaches work with the structure of the data and will result in usable information.
8	Evaluate current models and make appropriate changes to reduce bias and increase accuracy. Fine-tune hyperparameters to improve the fit of the models.
9	Meet with Primary Stakeholder to go over current findings and see if there are any

	other angles to pursue. Draft the poster utilizing the feedback from the primary stakeholder about what is most useful and important. Plan out future work for the next 3 weeks based on meeting feedback. Risk: The data is not revealing any useful information, or the primary stakeholder desires something in a completely different direction.
10 (M3 Poster Draft)	Complete the poster draft with the updated models and visualizations. Ensure that the poster is clear for all audiences, with enough information for technical audiences. Begin any work needed to pursue angles brought up in the week 9 meeting with the primary stakeholder.
11	Review all past work, evaluate, and modify if needed. Begin writing the writeup. Use any feedback from the poster to begin refining the poster. Use feedback about the poster to influence the writeup.
12 (M4 Writeup Draft)	Complete the write-up draft. Meet with the primary stakeholder with any updates or new information from the last meeting. Utilize any information from the meeting to influence writeup. Potentially have another deliverable for the primary stakeholder.
13	Utilize feedback to edit writeup and poster. Ensure that the poster and writeup meet the needs of the primary stakeholder and are tailored for their respective audiences.
14 (M5 final)	Final edits of poster and paper. Complete 2 days before the deadline to allow for a final look with fresh eyes.

10. DS3 Process and Tooling

- **GitHub Repo:** Github Repository URL
- **GitHub Project Board:** Github Projects Board URL
- **Discord:** Category #project-10 confirmed for team; peer project categories "joined"
- **Charter.md**: Committed.
  Vision: *Willamette University Undergraduate Admissions Office and enrollment management team will be able to identify, at the point of application, where a non-enrolled admitted student is likely to enroll instead. This will enable the Admissions Office to tailor materials and enable more targeted and effective recruitment strategies.*
  Mission: *The owner team will utilize student information, including demographics and the school where a student attended, to develop a classification and clustering model to predict what type of school (Willamette, private NW5 institution, or large university) an admitted applicant to Willamette University is likely to attend. The final deliverable will also include a pipeline that appends a predicted application type to each record in Slate.*
- **Backlog**: Seeded with milestone-tagged PBIs (should be 5-10)

- **Studio Loop**: After our studio session, we will respond to Critiques by 6 PM Thursday and capture follow-up items into the backlog, deliver an Iteration Review in README.md by Saturday at 12 PM, and respond to peer Briefs by Monday at 6 PM, with a first-pass adopt/defer/decline call for each item.